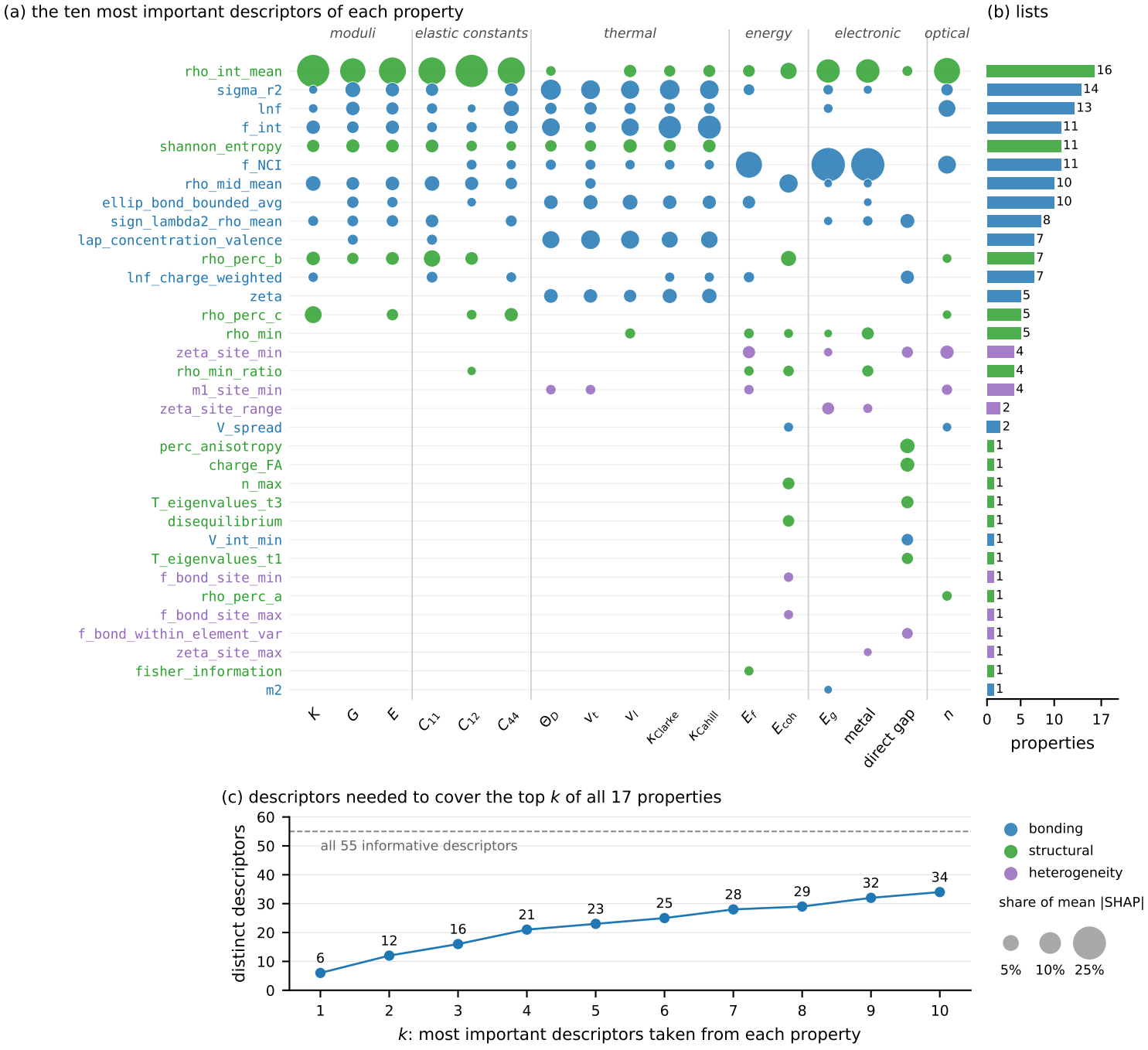

(a) the ten most important descriptors of each property
moduli
elastic constants
thermal
energy
electronic
optical
rho_int_mean
sigma_r2
lnf
f_int
shannon_entropy
f_NCI
rho_mid_mean
ellip_bond_bounded_avg
sign_lambda2_rho_mean
lap_concentration_valence
rho_perc_b
lnf_charge_weighted
zeta
rho_perc_c
rho_min
zeta_site_min
rho_min_ratio
m1_site_min
zeta_site_range
V_spread
perc_anisotropy
charge_FA
n_max
T_eigenvalues_t3
disequilibrium
V_int_min
T_eigenvalues_t1
f_bond_site_min
rho_perc_a
f_bond_site_max
f_bond_within_element_var
zeta_site_max
fisher_information
m2
K
G
E
C11
C12
C44
ΘD
vt
vl
κClarke
κCahill
Ef
Ecoh
Eg
metal
direct gap
n
(b) lists
16
14
13
11
11
11
10
10
8
7
7
7
5
5
5
4
4
4
2
2
1
1
1
1
1
1
1
1
1
1
1
1
1
0
5
10
17
properties
(c) descriptors needed to cover the top k of all 17 properties
distinct descriptors
all 55 informative descriptors
0
10
20
30
40
50
60
6
12
16
21
23
25
28
29
32
34
1
2
3
4
5
6
7
8
9
10
k: most important descriptors taken from each property
bonding
structural
heterogeneity
share of mean |SHAP|
5%
10%
25%